TIN SUM CHENG

Basel, Switzerland
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PROFESSIONAL SUMMARY

Postdoctoral researcher and machine learning scientist working across the full stack of foundation models: from model design to the kernel optimization in large-scale training. Publications at NeurIPS (×2), ICML, ICLR and TMLR. Industry experience at Huawei's Research Center. Regularly the bridge between ML and non-ML stakeholders, having worked alongside chemists and engineers on cross-disciplinary problems.

Core Competences

Programming Languages & Frameworks: Python, PyTorch, Triton (custom kernels)

ML Models: Kernel method, Language Transformers (Olmo, nanochat), Vision Transformer (ViT, DeiT), multimodal models (CLIP)

Infrastructure & tools: SLURM, multi-GPU training, Linux, Docker, Git, Weight and Biases

PROFESSIONAL EXPERIENCE

Postdoc at University of Basel, Switzerland | Oct 2026 – Present

- Uncertainty quantification for multimodal foundation models in pediatric healthcare

PhD in Computer Science (Machine Learning) at University of Basel, Switzerland | Feb 2022 – Sep 2026

- Pre-training and fine-tuning of large-scale foundation models on text (Nanochat), image (ViT), molecules (GNN), and multimodal (CLIP...)
- Designed reproducible ML experiment pipelines on HPC GPU clusters with Weights & Biases tracking and GitHub documentation

Research Intern at Huawei Hong Kong Research Center, Hong Kong | Aug 2025 – Jan 2026

- Optimized core matrix operators (MatMul, BMM) for next-generation hardware architectures using C and Triton, driving computational efficiency up to 60% in large-scale environments.
- Prototyped and integrated a neural network module into a legacy industrial system, improving algorithmic prediction accuracy by +75%.

EDUCATION

PhD in Computer Science (Machine Learning)
University of Basel, Switzerland | Feb 2022 – Sep 2026
Thesis: Spectral Analysis of Feature Embeddings in Modern Deep Learning: Theory of Optimization and Generalization
Supervisor: Prof. Dr. Aurelien Lucchi

MSc in Mathematics
University of Bonn, Germany | Mar 2019 – Sep 2021

BSc in Mathematics
The Chinese University of Hong Kong, Hong Kong | Sep 2014 – Sep 2018

SELECTED PUBLICATIONS

Attention Entropy Collapse Meets Loss Curvature: A Theory of Transformer Training Instability Foivos Alimisis, Tin Sum Cheng, Aurelien Lucchi *(Under review)*

Spectral Analysis of Molecular Kernels: When Richer Features Do Not Guarantee Better Generalization Asma Jamali, Tin Sum Cheng, Rodrigo A Vargas-Hernández *Transactions on Machine Learning Research (TMLR)*

Optimizer Choice Matters for the Emergence of Neural Collapse Jim Zhao, Tin Sum Cheng, Wojciech Masarczyk, Aurelien Lucchi *International Conference on Learning Representations* (ICLR 2026)

A Comprehensive Analysis on the Learning Curve in Kernel Ridge Regression Tin Sum Cheng, Aurelien Lucchi, Anastasis Kratsios, David Belius *Advances in Neural Information Processing Systems* (NeurIPS 2024)

Characterizing Overfitting in Kernel Ridgeless Regression Through the Eigenspectrum Tin Sum Cheng, Aurelien Lucchi, Anastasis Kratsios, David Belius *International Conference on Machine Learning* (ICML 2024)

A Theoretical Analysis of the Test Error of Finite-Rank Kernel Ridge Regression Tin Sum Cheng, Aurelien Lucchi, Anastasis Kratsios, Ivan Dokmanic, David Belius *Advances in Neural Information Processing Systems* (NeurIPS 2023)

LANGUAGES

English - Fluent (C1), German - Fluent (C1), Chinese - Native